
Pilot Corpus-Based Analysis of Emotional Vocabulary and Experiencer Constructions in Hittite

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Abstract. This paper presents a pilot corpus-based analysis of Hittite emotional vocabulary and experiencer constructions. It takes as its starting point the manually curated semantic annotations produced within the DFG-funded project *Akkadian and Hittite Emotion in Context* (AHEC), which were then extended through a combined automatic and philological workflow. The analysis is based on a working corpus of 2,906 annotated lexical-context records, expanded from an original set of 1,994 records. Additional material was retrieved directly from the digital corpora hosted by the Hethitologie-Portal Mainz (HPM) and from the TLHdig dataset published on Zenodo, then normalized and subjected to manual philological review. In total, the expansion added 912 records (45.7%) to the original corpus. The distributional analysis covers all 2,906 records. The genre-by-field contingency table yields $\chi^2(64) = 1261.77$, Monte Carlo $p < .00001$, and Cramér's $V = .233$; cluster-aware robustness checks confirm that the association is not reducible to repetition within a small set of texts. A preliminary constructional analysis then contrasts subject-experiencer and oblique-experiencer patterns. The study argues that quantitative distributional mapping is a prerequisite for controlled grammatical analysis in a corpus language.

1. Background and research questions

Emotional concepts in cuneiform cultures have been at the centre of scholarly attention for the past decade. For Hittite, the principal literature includes Galhano (2023) on the lexical and semantic organization of Hittite emotion vocabulary, Sonik and Steinert (2023) on discussion of emotional terms in Akkadian and Hittite, and the comparative Anatolian lexical dataset (Molina & Uvarova, 2026). The wider information on sensory and embodied contexts is represented in the *Routledge Handbook of the Senses in the Ancient Near East* (Neumann & Thomason, 2022), with an Anatolian section on the role of the senses among the Hittites.

What is still missing from the literature is a statistically explicit account of how emotional lexicon is distributed across the Hittite corpus and how emotional events are encoded grammatically. Existing work has primarily been lexicographic and philological: it establishes what individual lexemes mean, reconstructs their semantics, and situates them within cognitive conceptions of affect. A corpus-wide quantitative model of emotional vocabulary or a systematic account of the argument structure of emotion predicates is still a desideratum.

These are related but distinct questions. The first is lexical and distributional; the second is constructional. The central methodological claim of the present study is that the first is a prerequisite for the

second: before experiencer constructions can be analysed in a corpus language, the relevant predicates must be identified and their textual distribution established. This paper presents a pilot distributional analysis and a preliminary constructional analysis.

The investigation addresses two explicit questions:

1. To what extent does the distribution of Hittite emotional vocabulary vary across textual genres?
2. Do recurrent patterns of experiencer coding constitute lexeme-specific valencies or more general form–meaning pairings?

The working hypothesis is that genres differ not merely in the frequency of emotional vocabulary but in the lexical fields and constructions they preferentially recruit.

2. Data and research workflow

2.1 The AHEC database

The material derives from the semantic database developed by the project *Akkadian and Hittite Emotion in Context – Towards a Lexicon of Emotions in Cuneiform Texts from ancient Mesopotamia and Asia Minor* (AHEC, DFG #495257771) (Akkadian and Hittite Emotional Concepts in Context, 2026; Deutsche Forschungsgemeinschaft, 2026). The source AHEC workbook contained 1,994 entries. The resulting dataset of this study contains 2,906 annotated lexical-context records after retrieval, normalization, contextual review, and integration of the additional data, which makes the present study’s own addition 912 entries, or 45.7%.

A database entry is an annotated occurrence in context, and not a unique lexeme, manuscript, or text. The primary analysis retains every entry of the expanded corpus, including fragmentary, uncertain, and duplicate-looking entries.

For the principal distributional analysis, the records are assigned to nine main lexical-field categories:

- ANGER;
- JOY / HAPPINESS;
- HOSTILITY / RESENTMENT;
- MERCY / COMPASSION;
- DISTRESS / GRIEF;
- FEAR / ANXIETY;
- LOVE / AFFECTION;
- OTHER CORE EMOTION;
- SECONDARY / NON-CORE.

The last category contains emotion-adjacent evaluative, relational, regulatory, bodily, and metaphorical material. It is analytically relevant but is not treated as a single core emotion macrodomain.

The seed inventory itself is not the product of automatic extraction. It was compiled manually from textual contexts by philologists working with critical editions, especially Maria Molina and Ksenia Uvarova. Each entry reflects a contextual assessment of a lexeme’s meaning. This labour-intensive

procedure gives the AHEC dataset its philological value, but it also limits the size of the manually curated sample.

2.2 Projection onto larger digital corpora

The automatic stage begins from this manually curated inventory. I constructed a rule-based search lexicon from normalized AHEC lemmas and attested wordforms, supplemented it with forms documented in standard Hittite dictionaries, critical editions, and the second edition of Hoffner and Melchert's reference grammar (Hoffner & Melchert, 2024), and projected it onto a substantially larger body of transliterated text. The search space includes the digital critical editions hosted by the [Hethitologie-Portal Mainz](#) (HPM) (Academy of Sciences and Literature Mainz, 2026) and the complete TLHdig Beta 0.3 dataset released on Zenodo (Müller et al., 2026).

The automatic projection returns candidate attestations, not automatically validated emotional contexts. Three sources of noise therefore require explicit treatment.

The computational component also draws on earlier work on Hittite corpus engineering, syntactic annotation, and the treatment of fragmentary or multiply encoded text. Particularly relevant precedents include Frantíková's account of tagging and search design for a digital Hittite corpus (Frantíková, 2020), Molina's corpus-annotation proposals and the Hittite clause corpus (Molina, 2016a, 2016b; Molina & Molin, 2016), Inglese's discussion of fragmentary-text annotation and the PROIEL pilot (Inglese, 2018; Inglese, Molina, & Eckhoff, 2018), Smirnov's undergraduate thesis on automatic morphological ambiguity resolution (Smirnov, 2024), and the recent conversion of large Hittite XML collections into machine-learning-ready formats (Yavasan & Gordin, 2025). I use these studies as methodological precedents for normalization, token and annotation-layer preservation, fragment handling, and the explicit separation of automatically retrieved candidates from philologically validated attestations; the present workflow remains rule-based and manually adjudicated rather than a trained emotion classifier.

2.2.1 Polysemy

Polysemy is a structural feature of the lexicon rather than an occasional irregularity. A form such as *idālu-* ranges from general evaluative meanings such as “bad” or “evil” to more specifically affective uses. A search hit does not count as an emotional context until its interpretation is supported by the surrounding passage. The wider relation between negative evaluation, divine epithets, and Hittite concepts of evil is illustrated by Puértolas Rubio's study of negatively characterized deities (Puértolas Rubio, 2024); Molina's public-facing overview of Hittite anger and malice provides an accessible survey of several lexemes also represented in the present corpus (Molina, 2025).

2.2.2 Sumerographic spelling

Sumerographic spelling must be normalized. Several high-frequency items are frequently written logographically, for example TUKU.TUKU for *kardimmiya-* and 𐎶𐎵𐎶𐎵 for *idālu-*. Unless syllabic and logographic spellings are mapped to the same analytical lemma, the resulting frequency profile is distorted.

2.2.3 Metaphor and metonymy

Metaphorical and metonymic expressions require a separate analytical layer. Hittite *ištanza(na)-* “soul, mind” (ZI) and *lāla-* “tongue” (EME), for example, are not themselves emotion terms. They function as vehicles or loci in emotional expressions. Treating them as direct members of a core emotion field would inflate the relevant counts. In the present stage, such material is either assigned to SECONDARY / NON-CORE or reserved for a later metaphor-focused analysis.

2.3 Conservative filtering, validation, and corpus expansion

The retrieval procedure was deliberately conservative. The earlier TLHdig search space contained 851,270 analysable Hittite word units. I projected 1,629 normalized safe search forms, representing 293 normal lexemes or phrasal predicates, over this material. The broad retrieval stage generated 29,037 raw emotion-fragment candidates. These raw hits were not treated as automatically valid emotional attestations: 18,335 logographic or Sumerographic candidates were separated into a dedicated review layer, while 9,678 heuristic or otherwise unsafe matches were removed from the principal review queue.

Precisely because the project began from a conservative lower-bound strategy, I retained only about one thousand new candidate occurrences for intensive philological review: 1,024 clean lexical candidates from an initial pool of 29,037 raw matches. The aim was not maximal recall, but a review set whose provenance and contextual interpretation could be audited. After contextual adjudication, duplicate and parallel auditing, lemma normalization, and integration, the working corpus reached 2,906 records. Relative to the 1,994-row source workbook, this is a net increase of 912 records, or 45.7%.

Table 1: Main stages in the expansion from the source AHEC workbook to the author’s working corpus. Candidate categories describe the retrieval pipeline and are not all accepted corpus records.

Stage or measure	Rows / units
Source AHEC workbook supplied to the author	1,994 populated rows
Earlier TLHdig search space	851,270 word units
Raw automatically retrieved fragments	29,037
Logogram / Sumerogram candidates separated for independent review	18,335
Heuristic or unsafe matches removed from the principal queue	9,678
Clean candidates retained for manual review	1,024
Author-expanded analytical corpus	2,906 records
Net corpus growth relative to the source workbook	+912 rows (+45.7%)

Candidate hits were evaluated against their immediate collocates, syntactic environment, formulaic context, genre, critical translation, and duplicate or parallel status. Homonyms and broad non-emotional evaluative uses were preserved in the audit trail but were not automatically interpreted as emotional contexts. The relation between the 1,024 clean review candidates and the net increase of 912 corpus rows is not one-to-one: some candidates were duplicates, parallels, uncertain readings, or already represented contexts, while accepted material also required consolidation and metadata normalization.

The resulting 2,906-record corpus should therefore be understood as a conservative, author-expanded research corpus rather than an exhaustive inventory of Hittite emotional language. The much larger automatic candidate pool strongly suggests that substantially more material remains to be recovered. Future work will refine and gradually broaden the retrieval rules, re-examine selected logographic and formulaic candidates, and extend the search beyond preselected lexical items. In particular, systematic experiencer-based and construction-based searches will be developed to identify clauses in which emotional interpretation emerges from argument structure, event structure, or recurrent syntactic frames rather than from an already known emotion lexeme. This expansion is an integral planned stage of the project, not an optional supplement.

2.3.1 The later Beta 0.3 update

The Beta 0.3 comparison is a later incremental update between versions of my expanded corpus; it should not be confused with the much larger expansion from the original 1,994-row AHEC workbook

to the 2,906-record working corpus. The searchable archive increased from 21,853 to 23,902 unique witnesses and from 851,270 to 902,070 analysable Hittite word units. Exact-form comparison produced 1,333 candidate hits in 237 witnesses. Manual review retained 18 additional contexts: 13 high-confidence candidates and 5 probable candidates. This yields a candidate-retention rate of 1.35% for that update, not a corpus-wide precision estimate.

3. Distributional results

The distributional stage shows that genre is the strongest structuring variable tested in the present analysis. I do not yet claim that genre outweighs period, manuscript date, text length, or scribal tradition in a single multivariable model. The more cautious conclusion is that genre produces the clearest and most robust association among the analyses completed so far.

The corpus was normalized into nine macro-genres:

1. Correspondence;
2. Divination / Oracle;
3. Historiography / Political Narrative;
4. Lexical / Catalogue;
5. Myth / Literature;
6. Normative / Administrative / Technical;
7. Prayer / Hymn;
8. Ritual / Cult;
9. Unclassified / Fragmentary.

3.1 Genre distribution

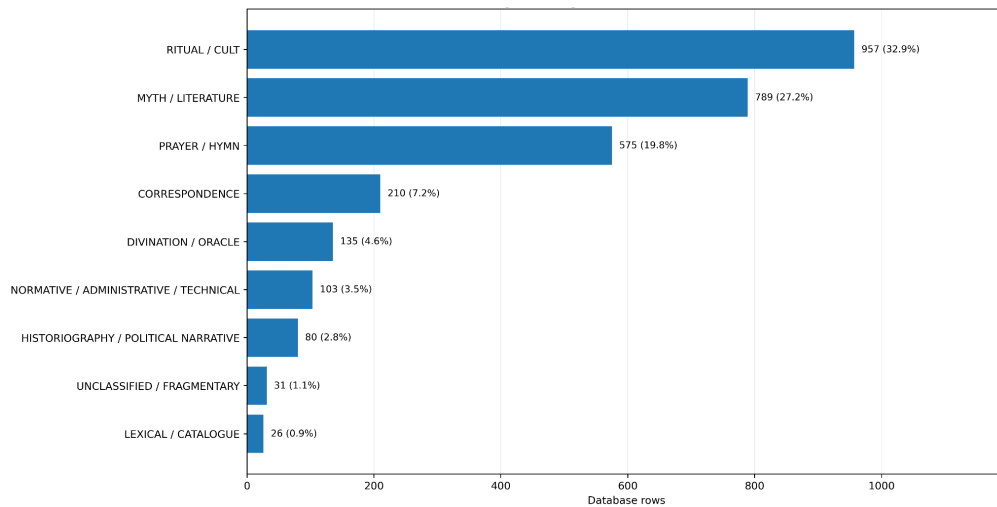
The database is strongly genre-imbalanced, but this imbalance is itself philologically meaningful. [Table 2](#) presents the full distribution.

Table 2: Distribution of the 2,906 database records across normalized macro-genres.

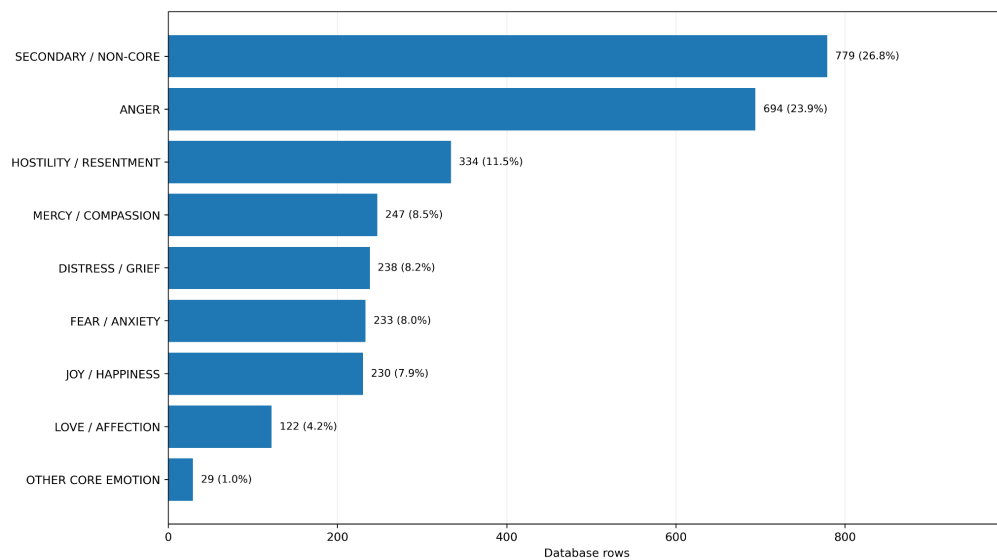
Macro-genre	Records	Share
Ritual / Cult	957	32.9%
Myth / Literature	789	27.2%
Prayer / Hymn	575	19.8%
Correspondence	210	7.2%
Divination / Oracle	135	4.6%
Normative / Administrative / Technical	103	3.5%
Historiography / Political Narrative	80	2.8%
Unclassified / Fragmentary	31	1.1%
Lexical / Catalogue	26	0.9%
Total	2,906	100.0%

Ritual / Cult, Myth / Literature, and Prayer / Hymn together account for 2,321 of the 2,906 records, or

79.9% of the database. This concentration reflects the communicative importance of emotional evaluation, divine-human interaction, conflict, lament, ritual regulation, and persuasion in these genres.



(a) Database records by normalized macro-genre.



(b) Database records by main lexical field.

Figure 1: Row-level distribution of the complete 2,906-record database. The unit represented is the annotated database record, not a unique text or token.

Figure 1 makes the two principal imbalances visible at a glance: the concentration of the database in ritual, mythological, and prayer material, and the numerical prominence of SECONDARY / NON-CORE and ANGER. These descriptive distributions motivate the contingency-table analysis but do not by themselves establish genre-conditioned selection.

3.2 Genre-conditioned lexical-semantic profiles

Genres differ not only in the quantity of emotional vocabulary they contain but also in its lexical-semantic composition. Correspondence is exceptionally rich in LOVE / AFFECTION: 73 of 210 records, or 34.8%, compared with 4.2% in the database as a whole. Myth / Literature strongly favours ANGER, with 323 of 789 records (40.9%), whereas Prayer / Hymn contains only 56 ANGER records out of 575 (9.7%). Prayer / Hymn instead favours MERCY / COMPASSION, with 120 of 575 records (20.9%), compared with 83 of 957 in Ritual / Cult (8.7%). Historiography / Political Narrative favours HOSTILITY / RESENTMENT,

with 22 of 80 records (27.5%). Divination / Oracle is dominated by ANGER, 78 of 135 records (57.8%), and DISTRESS / GRIEF, 34 of 135 (25.2%).

The overall genre-by-lexical-field contingency table departs strongly from independence: $\chi^2(64) = 1261.77$, Monte Carlo $p < .00001$, and Cramér's $V = .233$. A text-cluster bootstrap gives a 95% confidence interval of [.213, .266]. A cluster-aware permutation test remains significant at $p = .00010$, and a text-level presence/absence analysis still yields $V = .197$. The first two dimensions of the correspondence analysis account for 78.5% of total inertia. The association is therefore not reducible to repeated rows from a small number of texts. It is best interpreted as a robust genre-conditioned difference in lexical-semantic profile, not as direct causal influence of genre on lexical choice. The categorical analysis follows standard treatments of contingency tables and effect sizes (Agresti, 2019; Bulmer, 1979).

The combinations that contribute most visibly to the association are Correspondence $\times$ LOVE / AFFECTION, Myth / Literature $\times$ ANGER, Prayer / Hymn $\times$ MERCY / COMPASSION, Historiography / Political Narrative $\times$ HOSTILITY / RESENTMENT, and Divination / Oracle $\times$ ANGER. Prayer / Hymn $\times$ ANGER is correspondingly underrepresented.

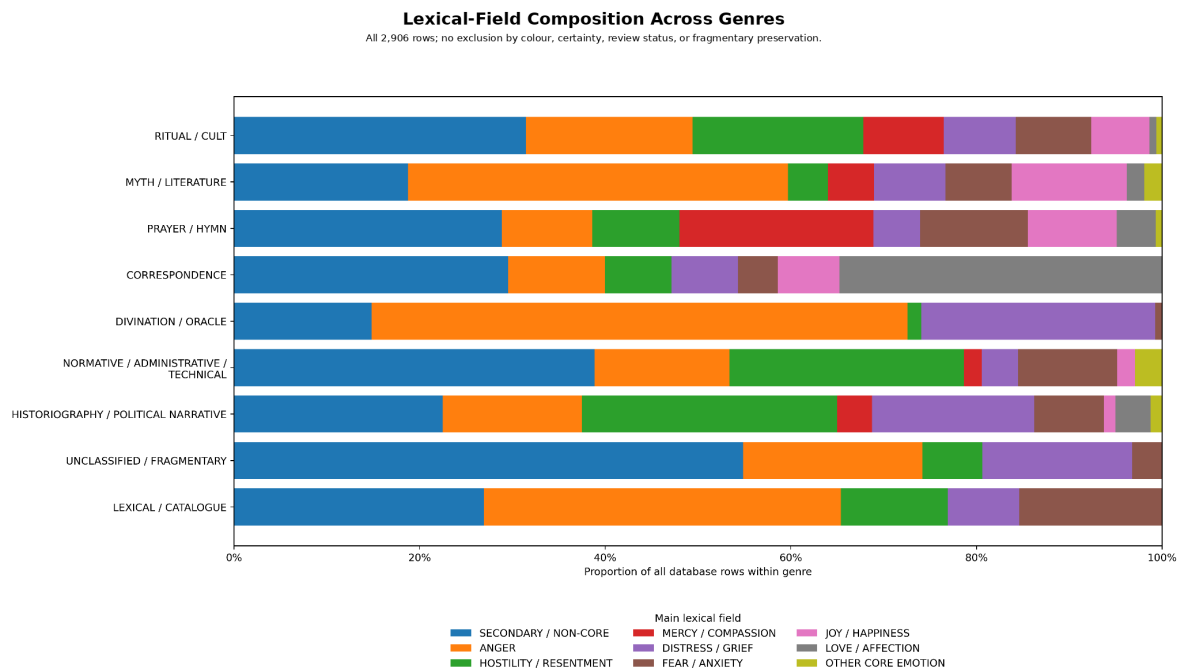
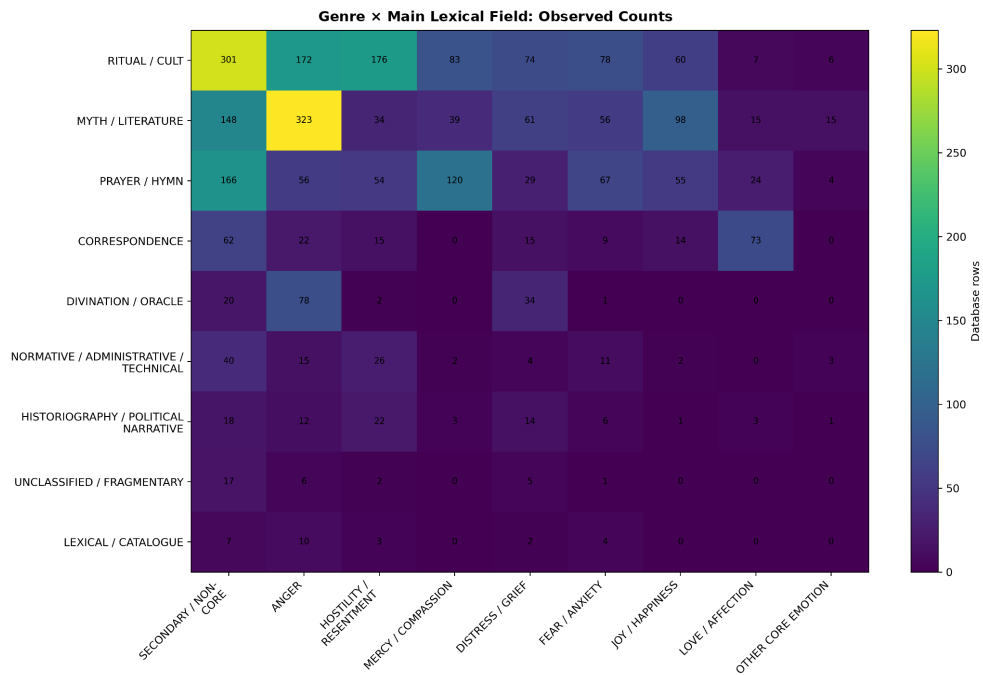


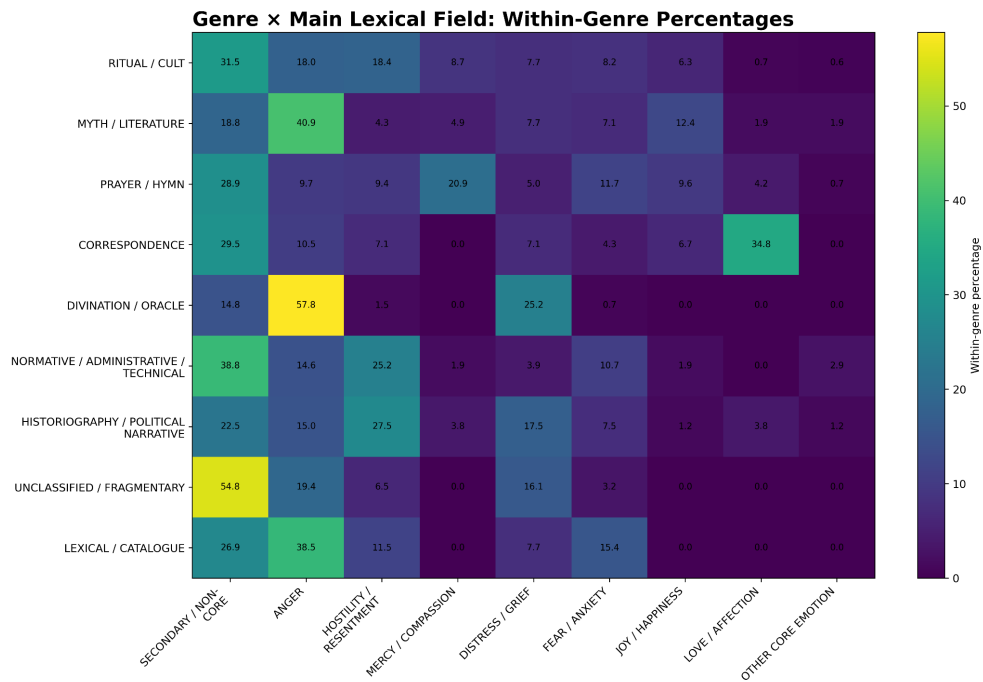
Figure 2: Within-genre composition of the nine main lexical-field categories. Each horizontal bar sums to 100% within a macro-genre, so the figure emphasizes compositional contrast rather than differences in genre size.

The stacked composition plot in Figure 2 and the paired heatmaps in Figure 3 separate three questions that raw totals can otherwise conflate: how large each genre is, how its internal lexical profile is composed, and which combinations are unusually frequent relative to independence. The adjusted-residual map in Figure 4 is therefore the most direct diagnostic of the cells responsible for the inferential result.

The correspondence-analysis map in Figure 5 summarizes the same association geometrically. It places genres near the lexical fields that contribute most strongly to their profiles, while preserving the caution that spatial proximity represents a distributional association rather than a direct historical or pragmatic cause.



(a) Raw database-row counts.



(b) Within-genre percentages.

Figure 3: Complementary views of the genre-by-lexical-field table. Raw counts show where the bulk of the annotated material lies; row percentages show which lexical fields are relatively characteristic of each genre.

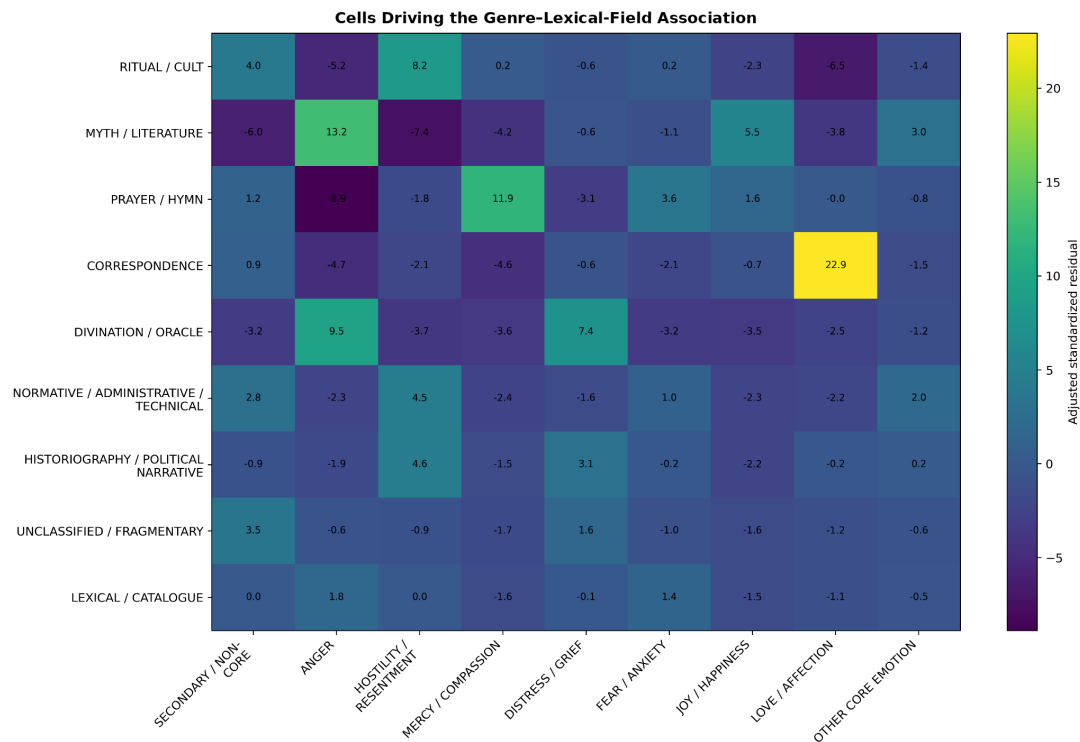


Figure 4: Adjusted residuals for the genre-by-lexical-field contingency table. Positive cells are observed more often than expected under independence; negative cells are underrepresented. This visualization identifies the combinations that drive the global χ^2 association.

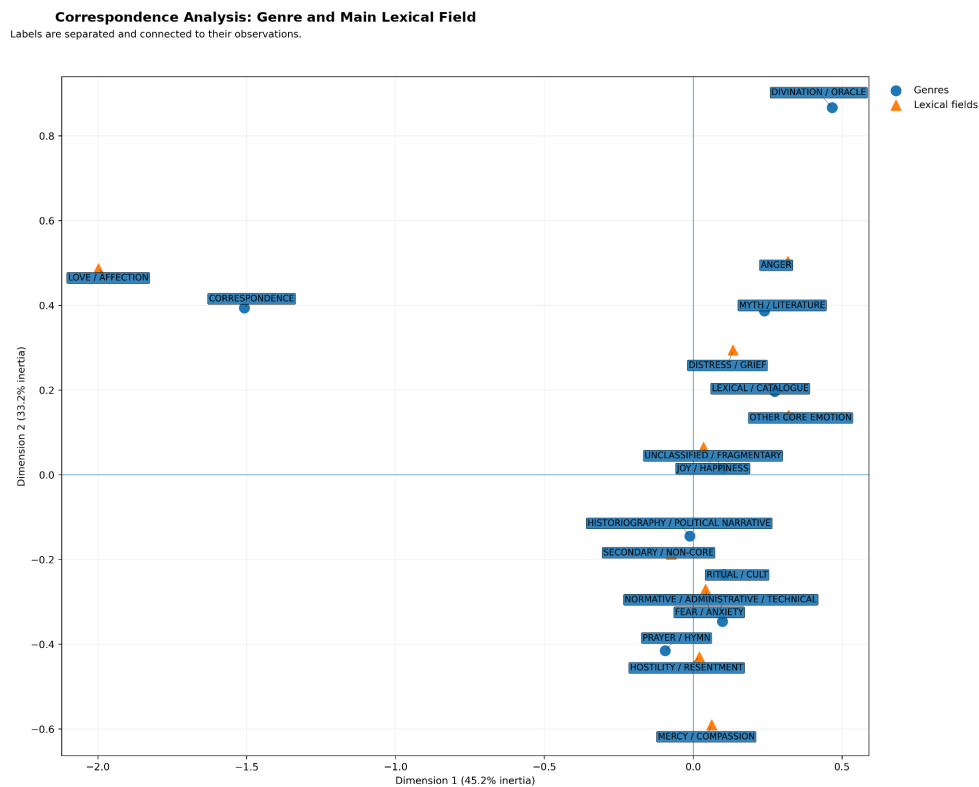


Figure 5: Correspondence analysis of genres and main lexical fields. The first two dimensions account for 78.5% of total inertia. Proximity represents association in the contingency-table geometry, not semantic identity or causal influence.

3.3 Lexical-field totals

Table 3 gives the row-level totals for the nine main lexical fields.

Table 3: Main lexical-field distribution in the complete analytical database.

Lexical field	Records	Share
SECONDARY / NON-CORE	779	26.8%
ANGER	694	23.9%
HOSTILITY / RESENTMENT	334	11.5%
MERCY / COMPASSION	247	8.5%
DISTRESS / GRIEF	238	8.2%
FEAR / ANXIETY	233	8.0%
JOY / HAPPINESS	230	7.9%
LOVE / AFFECTION	122	4.2%
OTHER CORE EMOTION	29	1.0%
Total	2,906	100.0%

3.4 Canonical-lemma concentration

The lexical-field totals are produced by a markedly uneven lemma-frequency distribution. A relatively small group of canonical lemmas supplies a large proportion of the database rows, while the remainder forms a long tail of less frequent items. Figure 6 displays the most frequent canonical lemmas in the expanded corpus. This concentration is one reason why the cluster-aware and presence/absence analyses are essential: raw row counts must not be interpreted as if every row represented an independent lexical innovation.

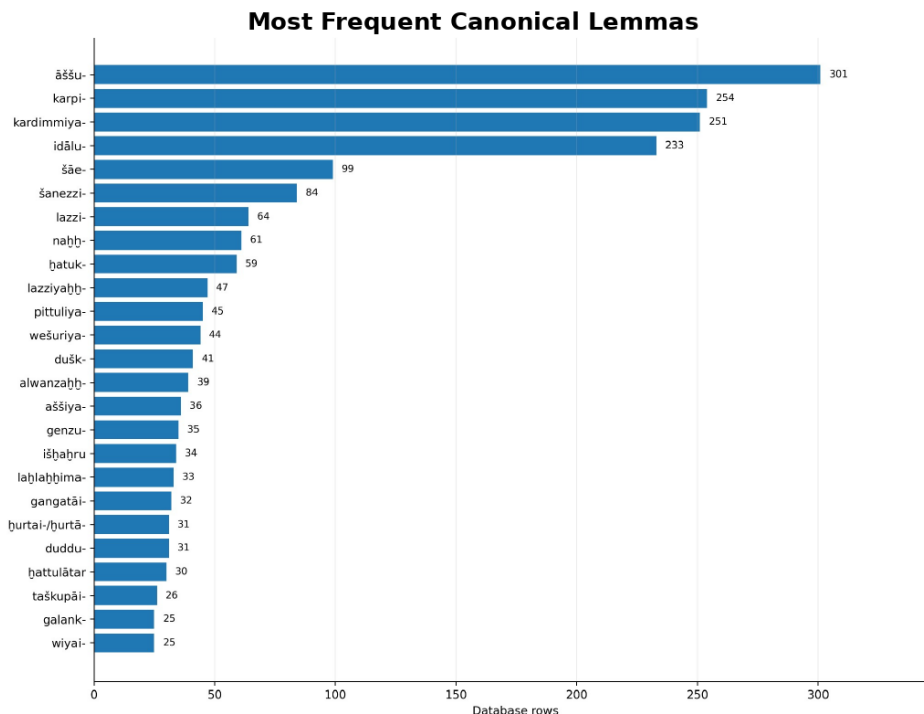


Figure 6: Most frequent canonical lemmas in the expanded database. Counts represent annotated database rows and therefore include repeated attestations and parallel or duplicate-looking records retained in the primary analysis.

3.5 Sensitivity analysis of secondary material

The primary analysis keeps SECONDARY / NON-CORE as a separate category. An exploratory semantic-affinity redistribution can nevertheless be used to examine what would happen if clearly emotion-adjacent material were reassigned to the nearest core field. As shown in Figure 7, this procedure increases especially JOY / HAPPINESS and HOSTILITY / RESENTMENT, but 104 records remain unassigned or non-emotional. The result is a sensitivity analysis rather than a revised primary classification.

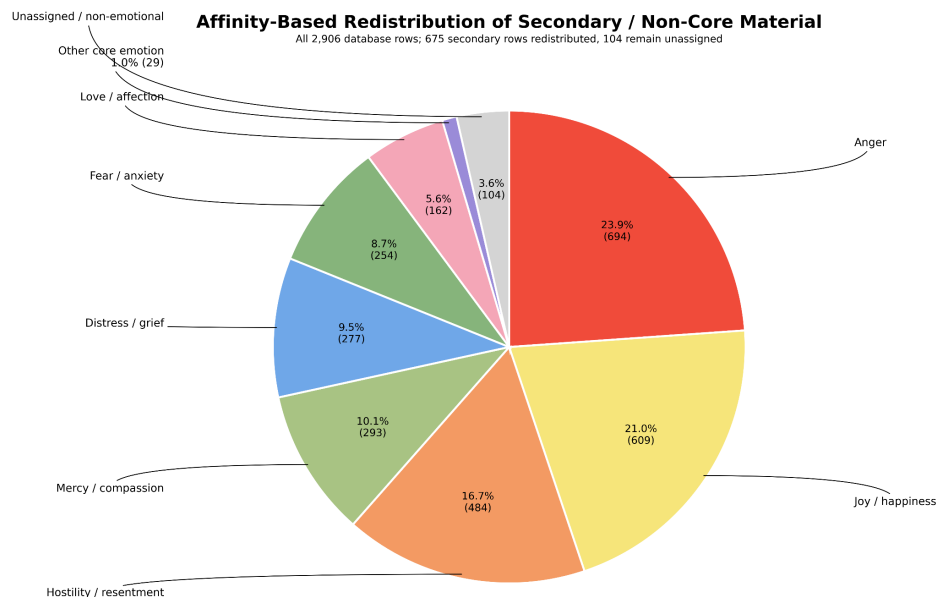


Figure 7: Exploratory affinity-based redistribution of SECONDARY / NON-CORE material. Of the 779 secondary rows, 675 are conservatively assigned to a core lexical field and 104 remain unassigned or non-emotional. Percentages are calculated over all 2,906 records.

Where TLHdig denominators are available, raw counts can be supplemented by exploratory rates per 1,000 analysable word units:

$$\text{rate per 1,000 words} = \frac{\text{emotion records}}{\text{analysable word units}} \times 1000.$$

These rates are treated as a secondary analysis because corpus coverage is not uniform across all CTH groups. They should not replace the primary row-level analysis.

Figure 8 demonstrates that high raw frequency and high normalized density are not identical. The figure is deliberately restricted to CTH groups for which a meaningful TLHdig denominator can be estimated and should therefore be read as a supplementary diagnostic rather than a population-wide ranking.

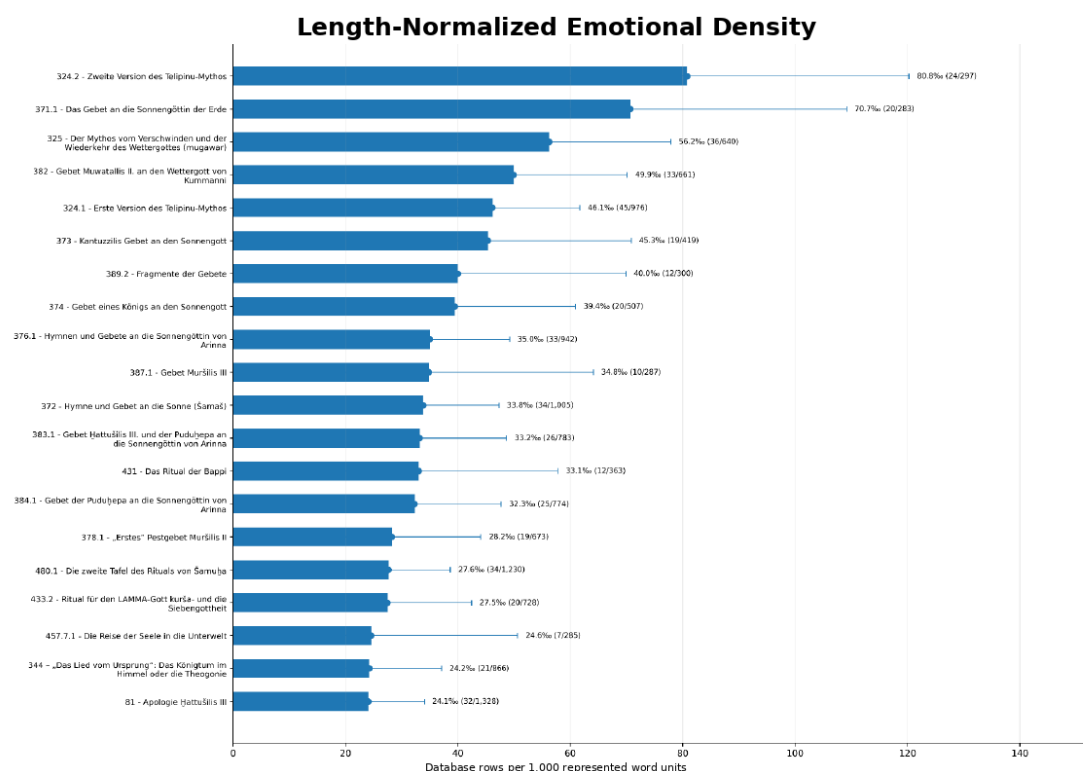


Figure 8: Length-normalized emotional density for selected CTH groups represented in TLHdig. Bars show database rows per 1,000 represented word units; the annotations also report the proportion and raw numerator/denominator used in the exploratory calculation.

4. Corpus overview and reproducibility

The [project repository](#) contains the analysis-ready table, normalization dictionaries, R scripts, contingency tables, density tables, and figures (Pankin & Molina, 2026). The current primary dataset is the author's expanded 2,906-record working corpus. It contains nine macro-genres, nine main lexical fields, 27 detailed lexical fields, 230 canonical lemmas, 166 lexical derivations, 646 distinct non-empty CTH designations, and 1,056 distinct publication sigla.

Lexical fields and lexical derivations are kept analytically separate: a lexical field is a semantic grouping, whereas a lexical derivation is a formal word-formation relationship between a base lexeme and its derivatives.

A complementary genre-level diversity view is given in [Figure 9](#). Lexical-field richness measures how many detailed fields are represented in a genre, whereas Shannon evenness measures how evenly its records are distributed among those fields. The figure shows that richness and evenness are distinct: a genre may cover many fields while still being dominated by a small number of them, as in Myth / Literature, or may have low richness but a comparatively even internal distribution.

The statistical and visualization workflow was implemented in R (R Core Team, 2026); text-processing and graphic design drew on reproducible approaches described by Silge and Robinson (2017) and Chang (2019). The quantitative results are therefore reproducible, but their interpretation remains philological.

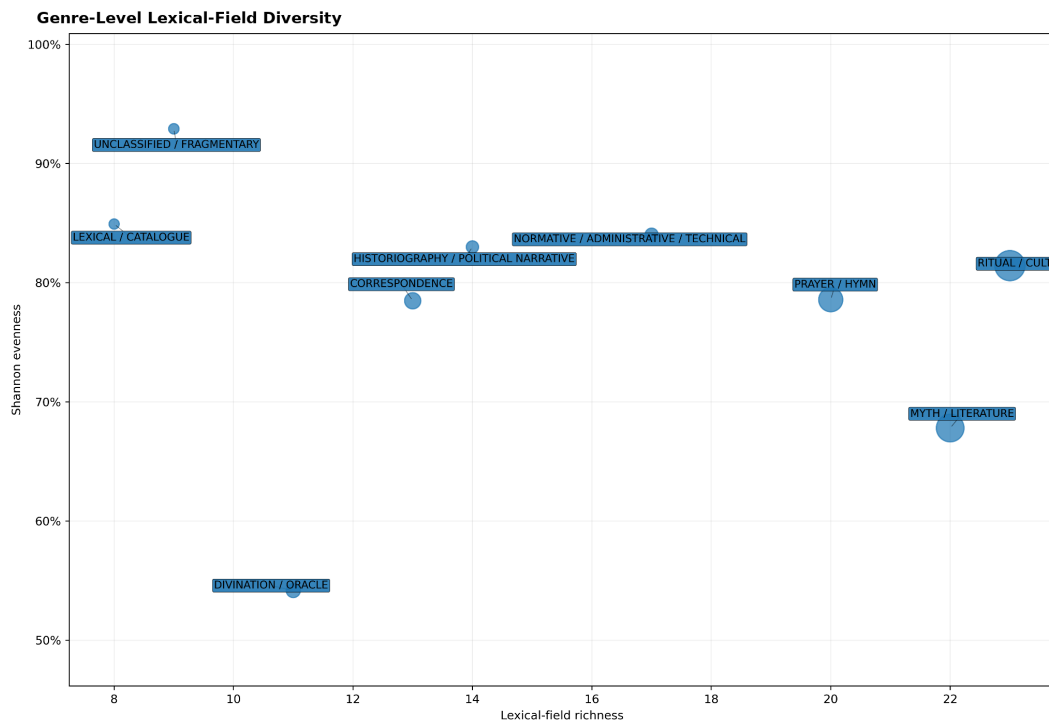


Figure 9: Genre-level lexical-field richness and Shannon evenness. Point size reflects genre size. The plot is descriptive and is not used as a substitute for the contingency-table analysis.

5. Experiencer constructions

The second stage concerns the grammatical encoding of emotional events: how the experiencer, stimulus or target, and causal participant are distributed across argument structure, and whether the observed regularities belong only to individual lexemes or also to recurrent constructions.

The typologically relevant contrast is between two broad encoding strategies (Luraghi, 2010). In the first, the experiencer is the grammatical subject: a nominative experiencer combines with an emotion predicate, while the target or cause may be expressed in an oblique phrase. The emotional state is construed as a state, property, or activity of the experiencer.

In the second, the emotion or stimulus is the grammatical subject and the experiencer is oblique. The emotional event is construed as something that arises in, becomes relevant to, or affects a participant. The experiencer is thereby profiled as the affected locus of an event rather than as a controlling agent.

Experiencers are typically animate, usually human or divine, and combine low control with high affectedness. Hittite nevertheless frequently codes them as nominative subjects, especially with transitive predicates of perception and cognition and with many intransitive or middle emotion predicates, including *karpiya-* and *kardimmiya-* “be angry,” *laḫlaḫḫiya-* “be anxious,” and *dušk-* “rejoice.” Dative experiencers are well established with *aššiya-* “be dear or pleasant,” *pukk-* “be hateful,” and *nakkeš-* “be burdensome”; accusative experiencers occur less frequently in force-dynamic or impersonal patterns.

5.1 Illustrative examples

5.1.1 Subject-experiencer pattern

KUB 21.19 + KBo 52.17 ii 10–11, CTH 383.1, Prayer / Hymn

nu=kan mān^dUTU^{URU} Arinna GAŠAN=YA ANA INIM^f Danuḫepa šer TUKU.TUKU-išta kwitki

“If the Sun-goddess of Arinna, my lady, somehow became angry because of the matter of Danuḫepa.”

Case frame: Overt nominative experiencer + intransitive emotion predicate; the cause or target is overtly expressed by the ANA ... šer phrase.

5.1.2 Dative-experiencer pattern

KUB 9.31 rev. iii 39–40, CTH 394, Ritual / Cult

nu=wa=šši=kan ŠA AMĪLŪTI UZU pug(ga)taru namma

“May human flesh become hateful to her or him again.”

Case frame: Overt dative experiencer =šši + nominative stimulus UZU “flesh” + middle predicate *pukk-*. The divine experiencer is recoverable from the preceding discourse.

5.1.3 Emotion-subject pattern

KUB 8.55 + KUB 33.123 ii 9, CTH 341.III.1, Myth / Literature

nu=šši kartimmiyaz kišat

“Anger arose in him.”

Case frame: Nominative emotion noun *kartimmiyaz* + dative experiencer =šši + change-of-state predicate *kiš-*. The experiencer is pronominal but overt; the discourse identifies him as Enkidu.

Table 4: Preliminary constructional annotation of the three illustrative examples.

Lemma / expression	Predicate / frame	Experiencer	Other core argument	Genre
<i>kardimmiya-</i>	verbal / middle emotion predicate	NOM	cause: ANA ... šer	Prayer / Hymn
<i>pukk-</i>	verbal / middle	DAT	NOM stimulus	Ritual / Cult
<i>kartimmiyaz kiš-</i>	noun + change-of-state verb	DAT	NOM emotion	Myth / Literature

5.2 Constructional interpretation

The preliminary constructional claim is that Hittite has at least two recurrent form–meaning pairings for emotional events: a subject-experiencer construction that profiles emotion as the participant’s state or activity, and an oblique-experiencer construction that profiles the emotion or stimulus as an event affecting a low-control participant. This is a constructional hypothesis in the descriptive sense of recurrent form–meaning association; it is not yet a claim that a fully conventionalized construction has been demonstrated in every period and genre (Goldberg, 2006; Hilpert, 2014; Traugott & Trousdale, 2013; Ungerer & Hartmann, 2023).

Three considerations support a constructional rather than purely lexical interpretation.

1. The pattern is not predictable from lexical meaning alone. Opposite-valence predicates such as *aššiya-* “be dear or pleasant” and *pukk-* “be hateful” share a dative-experiencer / nominative-

stimulus frame. The *kardimmiya*-derivational base is represented both by subject-experiencer predicates and by the emotion-subject clause *nu=šši kartimmiyaz kišat* “anger arose in him.”

2. The frame contributes a construal of its own. Subject-experiencer coding gives the experiencer greater discourse prominence and presents emotion as a state or activity attributed to that participant. Oblique-experiencer coding foregrounds affectedness, eventive onset, and relative lack of control by profiling the emotion or stimulus as the clause’s central figure. The contrast need not change the basic truth conditions of the emotional situation.
3. The clause-level frame organizes participants that cannot be recovered from a dictionary gloss alone: causal or target phrases such as *ANA ... šer*, dative experiencers, and overt emotion or stimulus subjects. This is where a constructional account offers explanatory value beyond a purely lexical list of valencies.

The limitations are equally important. The distributional analysis covers all 2,906 records, but the constructional pilot is organized around 29 core experiencer-predicate families identified from Luraghi’s inventory and supplemented by a small number of additional candidates. Experiencer case has not yet been exhaustively coded for every token in the database. Moreover, the conservative lexical search was designed to minimize false positives and consequently sacrifices recall: additional emotional contexts almost certainly remain in the larger candidate pool. The currently analysed examples are concentrated in prayers, rituals, and myths, the same genres that account for 79.9% of the database. The evidence is therefore sufficient to formulate and illustrate a recurrent alternation, but not yet to test a corpus-wide construction-by-genre model or to generalize without qualification to Hittite as a whole. A future expansion will combine broader lexical retrieval with morphosyntactic searches for experiencer frames, including constructions whose emotional status is not signalled by a preselected emotion lexeme.

6. Contribution and conclusion

The contribution of the study is twofold. First, it provides a genre-sensitive quantitative account of the Hittite emotional lexicon based on a systematically annotated corpus subset. Second, it demonstrates that a philological dataset created for lexicographic and semantic description can be repurposed as the empirical basis for a constructional analysis. This broader methodological point extends beyond Hittite: historical-linguistic datasets are often richer than the questions for which they were originally assembled.

The central empirical result is not merely that some genres contain more emotion terms than others. Different genres select systematically different parts of the emotional lexicon in ways that correspond to their discourse functions. The central methodological result is that this distributional map makes a controlled constructional study possible.

Future diachronic modelling should keep composition date and manuscript date as separate variables rather than collapsing them into a single period label. Dating certainty, genre, and scribal tradition should likewise be represented independently. The aim is to model chronological structure without discarding most of the corpus or conflating composition history with manuscript transmission.

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Data and credit statement

The source AHEC workbook and its underlying manual semantic annotations were produced within the AHEC project and supplied to me under the supervision of Maria Molina. The 2,906-record working corpus analysed here is my expanded derivative corpus, developed from the 1,994-row source workbook through search-lexicon construction, HPM/TLHdig retrieval, normalization, candidate filtering, duplicate and parallel auditing, contextual philological review, genre and lexical-field aggregation, and integration. The statistical analysis, density calculations, visualizations, and preliminary constructional reinterpretation reported here were likewise carried out by the author. This distinction is essential: AHEC supplied the philological foundation, whereas the corpus expansion from 1,994 to 2,906 rows and the quantitative workflow are contributions of the present study.

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